SAMARTH

CUET PG 2027

Lecture – 08 | Computer Science

Scoring Topic

Unit 03 : Digital Fundamental

Topic : Boolean Algebra and Logic Gates: Part 1

Very Imp

Compulsory Q.

By- Puneet Kaur



Topics *to be covered*

1. Boolean Algebra and Logic Gates: Part 1



Topic : Boolean Algebra and Logic Gates: Part 1

0, 1 — bits.

0, 1

Boolean Algebra Part-1

Content:

1. Axioms Of B.A.
2. Some Useful Theorems
3. Unique Features
4. Boolean Subalgebra
5. Basic Operations
6. Logic Gates

A non-empty set B with two binary operations + and., a unary operation', and two distinct elements 0 and 1 is called a Boolean algebra denoted by (B, +,.,', 0, 1) if and only if the following properties are satisfied.

Topic : Boolean Algebra and Logic Gates: Part 1

LEARN

Axioms of Boolean algebra-

If a, b, c £ B, then

1. **Commutative laws:**
   - (a) $a + b = b + a$
   - (b) $a.b = b.a$

2. **Distributive laws:**
   - ~~(a) a + b = b + a~~
   - (b) $a.(b + c) = (a.b) + (a.c)$

LAWS

VIMP

① Direct Expected

② Makes the full.

③ Solve the question.

All laws are provable with truth table

a	b	a+b	b+a	a.b	b.a
0	0	0	0	0	0
0	1	1	1	0	0
1	0	1	1	0	0
1	1	1	1	1	1

④ $a + (b.c) = (a+b).(a+c)$

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3. identity laws:
   (a) ~~a + a′ =1~~
   (b) ~~a.a′ =0~~

$A + 0 = A$
$A \cdot 1 = A$

Additive identity = 0
Multiplicative identity = 1

A·1	1	A	0	A+0
0	1	0	0	0
1	1	1	0	1

4. complement laws:
   (a) a + a′ =1
   (b) a.a′ =0

a	a′	a + a′	a.a′
0	1	1	0
1	0	1	0

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Basic theorems-

Let a, b, c £ B, then

1. idempotent laws:
   (a) a +a =a
   (b) a.a.=a

Ex

a	a	a+a	a.a
0	0	0	0
1	1	1	1

2. **Boundedness laws:**
   (a) a + 1 =1
   (b) a.0 =0

a	1	a+1	0	a.0
0	1	1	0	0
1	1	1	0	0

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3. **Absorption laws:**
   (a) a + (a.b) = a
   (b) a.(a +b) =a

$= a + ab$
$= a(1 + b)$
$= a \cdot 1$
$= a$

$= a \cdot (a+b)$
$= \frac{aa}{a} + ab$
$= a + ab$
$= a(1+b)$
$= a \cdot 1$
$= a$

4. **Associative laws:**
   (a) (a + b) +c = a+(b+c)
   (b) (a.b).c =a.(b.c)

① $(a+b)+c = a+(b+c)$

② $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

a	a	a·a
0	0	0
1	1	1

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$(+)' = \cdot$

$(\cdot)' = +$

Same as complement law.

5. **Uniqueness of complement:**
   $a + x = 1$ and $ax = 0$, then $x = a'$

6. **Involution law:**
   a. $(a')' = a$
   b. $0' = 1$
   c. $1' = 0$

$(+)' = \cdot$

$(\cdot)' = +$

7. **De-Morgan's laws:** V. imp
   (a) $(a + b)' = a'.b'$
   (b) $(a.b)' = a' + b'$

① $(a+b)'$

$= a' \cdot b'$

② $(a \cdot b)'$

$= a' + b'$

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- Note: where there is no danger of confusion, the symbol can be deleted just as in writing algebraic products i.e., a.b as ab.
- Algebraic proofs of some laws are given and the rest are left as exercises for the readers.

Each of the laws can also be proved using a truth table.

Theorem : $a + a = a$

Theorem: $a.a = a$

Theorem: $a + 1 = 1$

Theorem : $a + (a.b) = a$

Theorem: for each $a \in B$, a' is unique.

Theorem: The element 0 and 1 are unique

Theorem: (i) $(a+b)' = a'b'$ (ii) $(a.b)' = a' + b'$

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Unique Feature:

- The striking unique features Boolean algebra, which distinguishes it from the other branches a algebra are:
  (a) Boolean algebra has only a finite set of elements, namely 1 and 0, which are usually called true and false, where as ordinary algebra deals with a set of an infinite number of elements. Thus all the variables and constant are allowed to have only two possible values 0 and 1. Hence it provides the operations and rules for working with the set {0,1};
  (b) Boolean algebra does not have operations equivalent to subtraction and division;
  (c) In ordinary algebra, there is no equivalent of the unary operation of bar or/over a variable known as complementing.

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Boolean Sub algebra:

- A non-empty subset A of a Boolean algebra (B, +,. , 0, 1) is called a Boollean sub algebra if A is a Boolean algebra under the same set binary and unary operations i.e. a, b ∈ A a+b, a, b, d ∈ A.
- For example, consider a set D30 = {1, 2, 3, 5, 6, 10, 15, 30}, the divisor of 30 and let +,. And be defined on D30 as follows.

  a+b = 1cm (a,b)

  a.b = gcd (a,b) and a1 = 30/a

- then D30 is a Boolean algebra and A1 = {1, 5, 6, 30} and A_3 = { 1, 2, 3, 6} are Boolean sub algebra of D30 since A_1, A_2 and A_3 are closed under +,. But A_4 ={1, 2, 3, 6} is not a Boolean sub algebra because 6 = 30/6=5 does not belong to A_4. To show that A is a Boolean sub algebra it is not necessary to verify the closureness with respect to all three operations. It is enough to show that A is closed under the set of operations {+, /} or {., /}.

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Basic operations-

- The basic operations used in Boolean algebra are logical addition, logical multiplication and complementation. These basic operations are called logical operations. Boolean 1 and 0 do not represent actual numbers. In the digital logic field several terms are used continuously with 0 and 1.

Logic 0	False	Off	Low	Open	No
Logic 1	True	On	High	Closed	Yes

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Logical addition (OR operation)

- Each variable in Boolean algebra has either two values: true or false (1 or 0), the operation can be defined as follows-

0 +0 = 0
0 + 1 = 1
1 + 0 = 1
1 + 1 = 1

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

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- Where logical addition is denoted by + or by OR. if we write C= A+B, one can determine C by listing all possible combinations for A and B in a tabular form which is known as truth table of logical addition.

Input		Output C = A + B
A	B	
0	0	0
1	0	1
0	1	2
1	1	1

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- The symbol + does not have the normal meaning, but it is a logical addition and is referred to as OR operation. The equation A + B = C can be read as AOR B equals C. this concept can be extended to any number of variables. To avoid ambiguity, a number of other symbols have been recommended as replacement for + sign.

Logical multiplication (AND operation)-

The logical multiplication can be defined as follows

0.0 = 0

0.1 = 0

1.0 = 0

1.1 = 1

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

- Where logical multiplication is denoted by, or by AND. If we write C = AB, we can determine C by listing all possible combination of A and B. The truth table for logical multiplication of two variables.

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Input		Output C = A . B
A	B	
0	0	0
1	0	0
0	1	0
1	1	1

It is apparent from the table that the AND operation is exactly the same as ordinary multiplication. The equation A.B = C can be read as A AND B equals C.

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$\bar{0} = 1$

$\bar{1} = 0$

Complementation- (NOT)

- In Boolean algebra, an operation called complementation is used for which symbol is (bar over)' or' and this can be defined as 0' = 1 and 1' = 0.
- Here A' means the complement of A and read as NOT A. It is opposite to the logic value of A=1.

$1! = 0$

$0! = 1$

Input	Output
A	B = A'
0	1
1	0

Duality-

Dual.

- Given a Boolean expression, the dual is formed by replacing AND with OR, OR with AND, O with 1, and 1 with 0. Variables and complements are left unchanged. This rule for forming the dual can be summarized as follows:
- $[f(X_1, X_2, \ldots X_N, 0, 1, +)]D = f(x_1, X_2, \ldots X_N, 1, 0, \ldots +)$

Rule :-

1. 0 change to 1
2. 1 change to 0
3. + change to .
4. . change to +

NOTE :- Variables & its complements remain unchanged.

Ex $F = (a+1)\cdot(b'\cdot 0)$

$F^D = (a\cdot 0)+(b'+1)$

Q:

$F = (1+x')\cdot a'$

$F^D = (0\cdot x')+a'$

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- This important property of changing operations and Identity elements in Boolean algebra is known as duality principle. This principle ensures that if a theorem is proved based on the axioms of the Boolean algebra, then dual theorem obtained by interchanging + with and 0 with I automatically holds. Axioms and basic theorems are divided into two parts (a) and (6), one part may be obtained by other by using duality principle. For example. If f= (a.1). (0.a') then fD = (a + 0) + (1 + a') which is obtained interchanging + and, and interchanging 0 and 1.

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De-Morgan's Theorem-

- The complement of any Boolean expression can be found by successively applying the theorem referred to as De Morgan's theorem. The two theorems are

$(x+y)' = x' y'$

$(xy)' = x' + y'$

x	y	x'	y'	x+y	(x+y)'	x'y'	xy	(xy)'	x'+y'
0	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	0	0	1	1
1	0	0	1	1	0	0	0	1	1
1	1	0	0	1	0	0	1	0	0

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- These theorems can be extended to more than one variable. For instance, consider (x+ y + z)'
  (x + y + z)' = (A + z)'
  = A' z'
  = (x+y)' z'
  = x' y' z'

Similarly, it can be proved (xyz)' = x' + y' z'
Proceeding similarly, we can extend the results to any number of variables.

- The theorems are equally valid where x and / or y are expressions that contain more than one variable.

For example,

$(AB' + C)' = (AB')' C'$

$= (A' + (B')') C'$

$= (A' + B) C'$

$= A'C' + BC'$

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- We can combine the two forms of De Morgan's theorem into a single rule which will enable us to complement an expression. To obtain the complement of a Boolean expression replace
  (a) 0 with 1
  (b) 1 with 0
  (c) + with *
  (d) . with +
  (e) Each variable with its complement.
  Which can be summarized as?
  $[f(X_1, X_2, \ldots X_N, 0, 1, +)]' = f(X_1', X_2', \ldots X_N', 1, 0, \ldots +)$.

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Logical Gate-

- A logic gate is an electronic circuit that operates on one or more input signals to produce an output signal. Gates are digital circuits and often called logic circuits. Gates are represented by symbols, the lines entering the symbol from left are inputs, and the line on the right is the output. Placing a small circle on an input or output complements the signal on that line.

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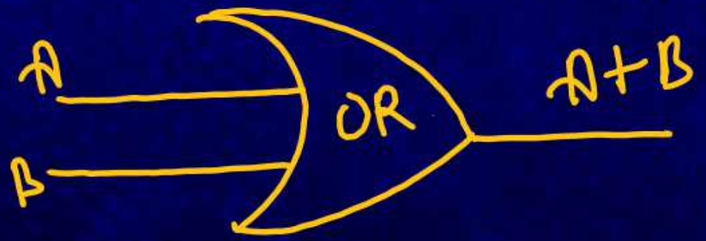
OR gate-

- The OR gate is an electronic device which receives two or more inputs and produces an output equal to the OR sum of the inputs. It is the symbol for two input OR gate and its truth table. An input signal applied to gate has only two stable states either 1 (high) or o (low) and output is I (high) if at least one of the inputs is high. The output is 0 if all inputs are 0.

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1



Symbol for an OR gate

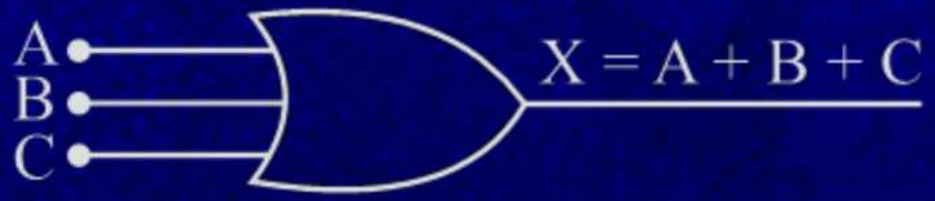


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Input		Output F = A + B
A	B	
1	1	1
1	0	1
0	1	1
0	0	0

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- The same idea can be extended to more than two inputs. A three input OR gate and its truth table.



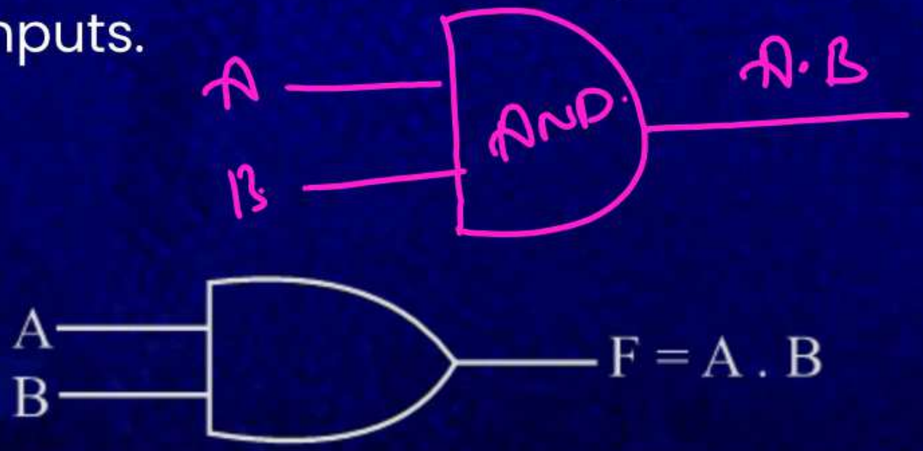
Symbol for these input
OR gate

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Input			Output
A	B	C	X = A + B + C
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

AND gate-

- AND gate is an electronic device that accepts the values of two or more Boolean variables as input and produces their Boolean sum as output. Same operation holds good for an AND gate with more than the two inputs.



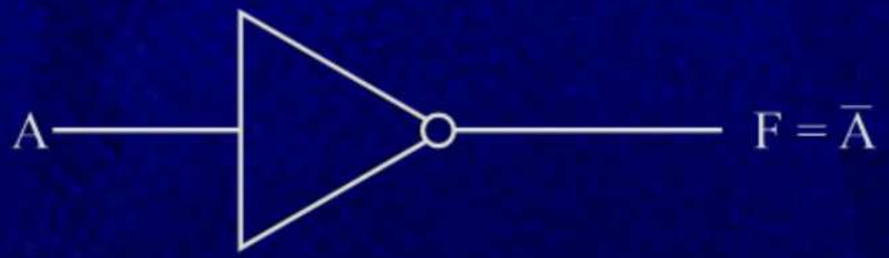
Symbol for an AND gate

Input		Output
A	B	F = A . B
1	1	1
1	0	0
0	1	0
0	0	0

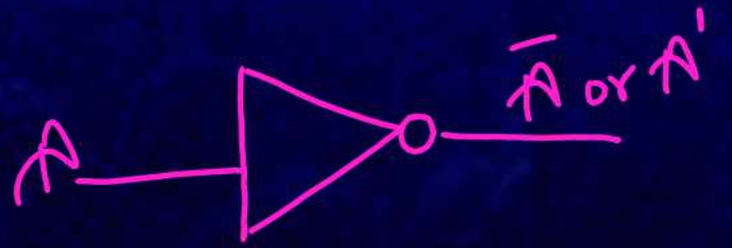
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NOT gate- Complement.

- A NOT gate has only one input and one output. It is also called an inverter. Its output is the complement of the input.



A	F = A'
1	0
0	1

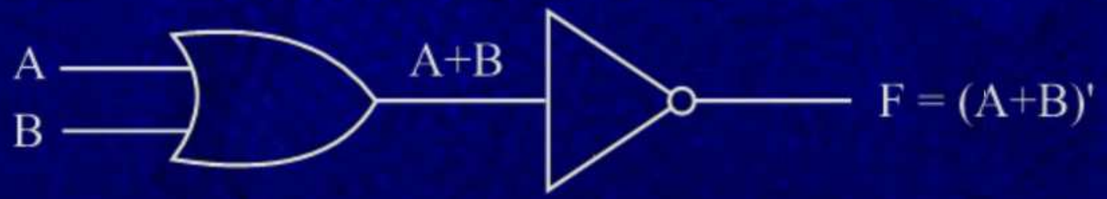


NOR Gate



NOR gate- NOT(OR)

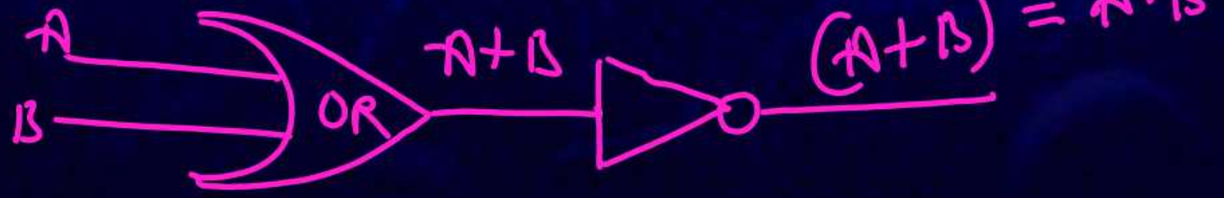
- This gate is logically equivalent to a NOT gate with an OR gate. The NOR function is the complement of OR function.



A NOT gate in series with an OR gate.

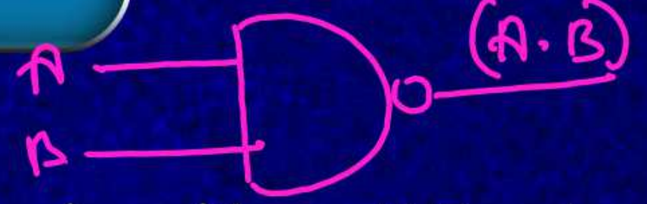


A	B	F = (A + B)'
1	1	0
1	0	0
0	1	0
0	0	1



$(A+B)' = A' \cdot B'$

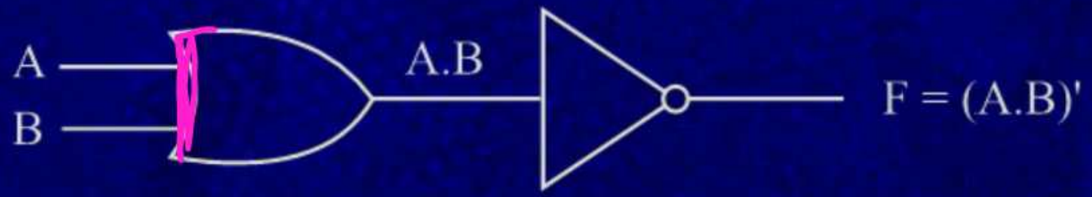
NAND →



NAND gate-

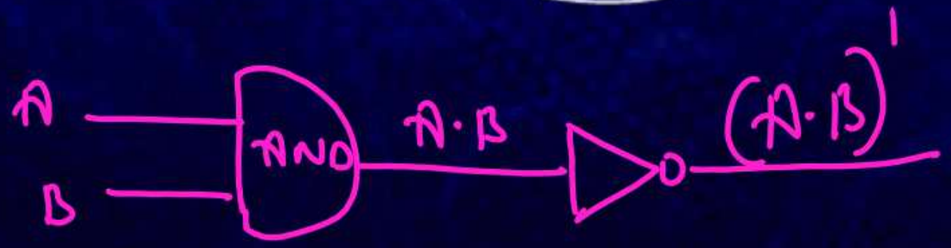
NOT (AND)

- This gate is logically equivalent to a NOT gate in series with an AND gate.



A NOT gate series with an AND gate

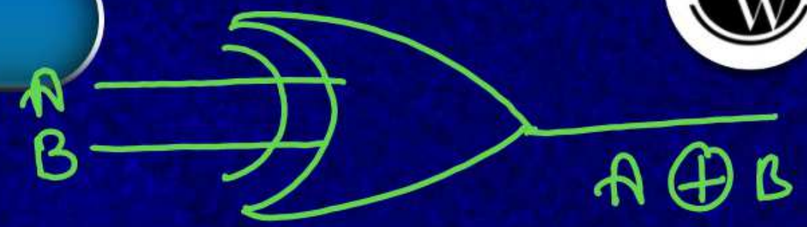




A	B	F = (A.B)'
1	1	0
1	0	1
0	1	1
0	0	1

$(A \cdot B)' = A' + B'$ A2

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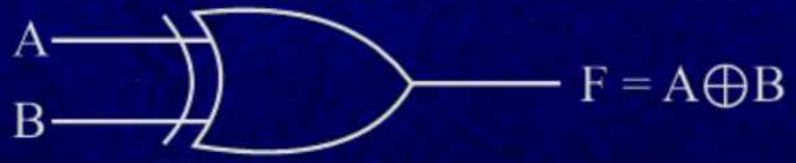


XOR

Exclusive OR gate- ⊕

- The full name of OR gate is the inclusive OR gate. The exclusive OR gate is different. It gives an output of 1 when either but not both inputs are I. In Boolean algebra, the sign stands for XOR operation.

$A \oplus B = A'B + AB'$



Symbol for a XOR gate

Input A	Input B	Output F = A'B + AB'
0	0	0
0	1	1
1	0	1
1	1	0

Rule

Same Input = O/P 0

Diff. Input = O/P 1

Exclusive – NOR (XNOR) gate-

(XOR)'

- XNOR gate is equivalent to an Exclusive OR gate followed by an inverter. The output is I only we both inputs are either 0 or 1.

Rule

Same Input ⇒ O/P 1

Diff Input ⇒ O/P 0

Input		Output
A	B	F = AB + A'B'
0	0	1 ✓
0	1	0
1	0	0
1	1	1 ✓

XNOR operation can be written as follows:

$F = AB + A'B'$

$= A \text{ XNOR } B$

$= (A \oplus B)'$

The logic diagram and logic symbol is shown in figure.



Topic : Boolean Algebra and Logic Gates: Part 1

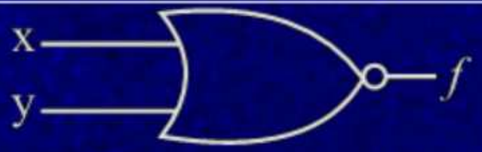
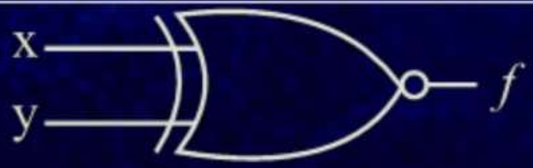
Summary Table

Logic Gates With truth Table:

Name	Symbol	Truth Table	Boolean Formula
AND		x y f 0 0 0 0 1 0 1 0 0 1 1 1	$f = xy$
OR		x y f 0 0 0 0 1 1 1 0 1 1 1 1	$f = x + y$
NOT (inverter)		x f 0 1 1 0	$f = x'$

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Universal gates

NAND		x y | f 0 0 | 1 0 1 | 1 1 0 | 1 1 1 | 0	$f = (xy)'$
NOR		x y | f 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 0	$f = (x + y)'$
XOR (Exclusive-OR)		x y | f 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 0	$f = x \oplus y = x'y + xy'$
XNOR (Exclusive-NOR)		x y | f 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 1	$f = (x \oplus y)' = xy + x'y'$

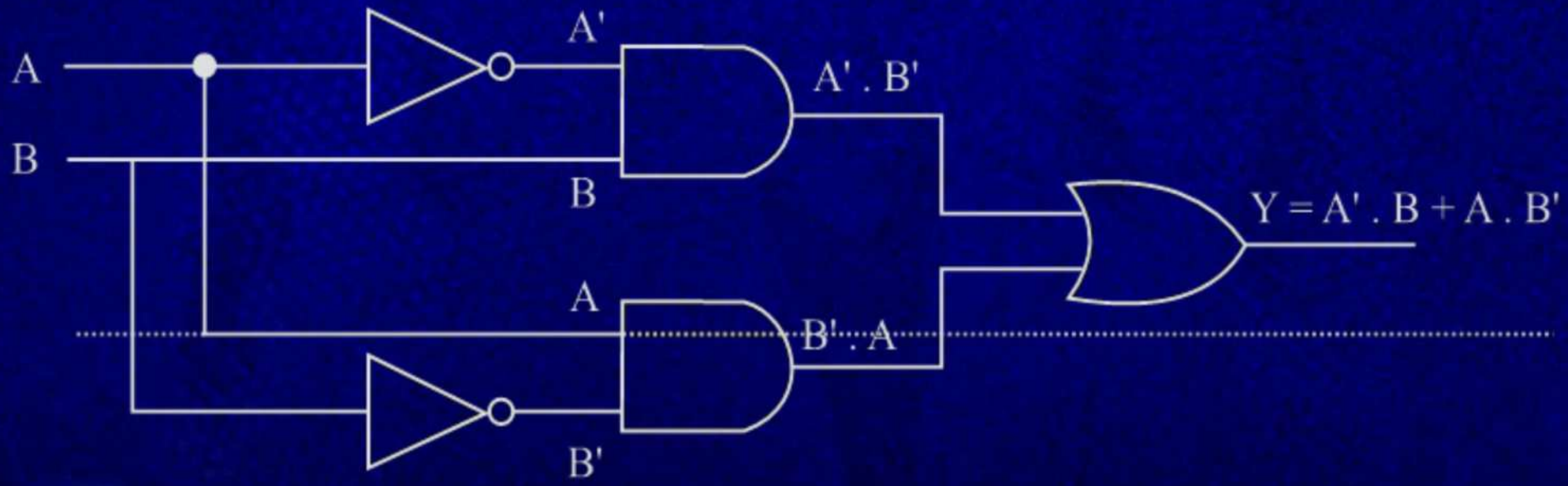
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Interconnecting gates-

Next semi

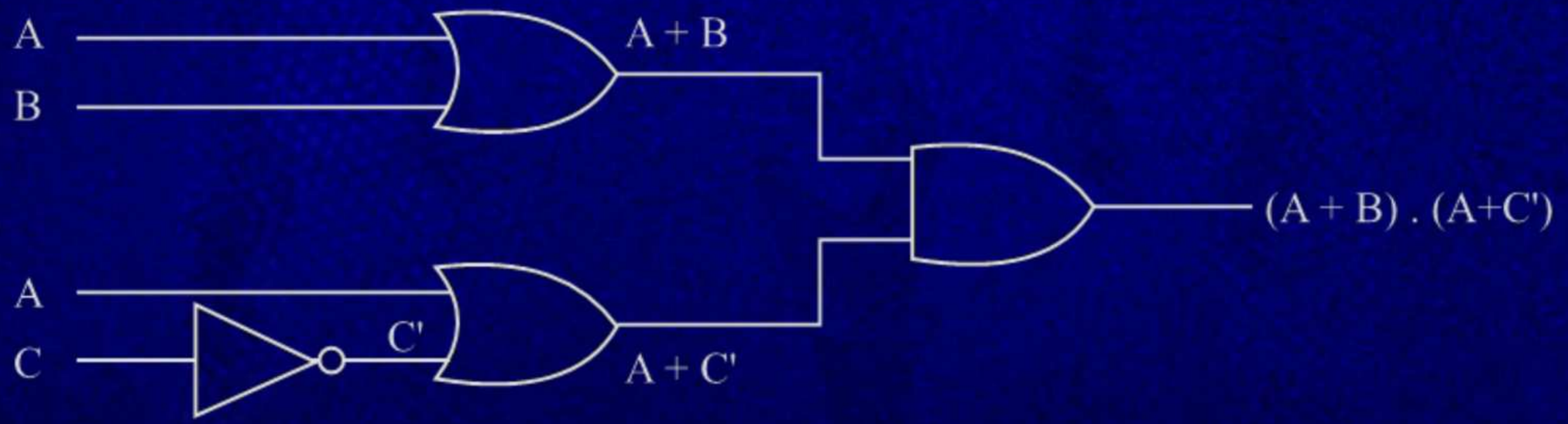
- The gates can be interconnected to form gating or log networks. The Boolean expression corresponding to given network, can be derived by systematically progressing from input to output on the gates. When gating network is formed, some gates may share inputs. One method is to indicate the inputs separately for each gate; the other method is to use branching that indicates all the gates that use a given input.

Topic : Boolean Algebra and Logic Gates: Part 1



(a)

Topic : Boolean Algebra and Logic Gates: Part 1



(b)

2 Mins Summary

Boolean Algebra + Gates

Basics

THANK YOU